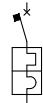
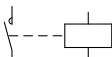
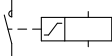


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TENDRARA PROJECT GAS TREATMENT AND LNG PRODUCTION PLANT ON RENTAL BASIS																					
SAFETY CRITICAL ELEMENT LIST																					
IFI	2	31/01/2025	AS-BUILT				ADF	GDP	IC												
IFI	1	06/12/2024	GENEERAL REVIEW				ADF	GDP	IC												
IFI	0	26/03/2024	ISSUED FOR COMMENTs				ADF	GDP	IC												
Status	Rev. n.	Rev. n.	Rev. n.				Rev. n.	Rev. n.	Rev. n.												
	Rev. Index	Date	Description				Prepared	Checked	Approved												
Client							Project Title														
ManaEnergy Managem Group MANAENERGY Mahaj Riad Center, Batiment 6, 3eme etage Avenue Al Arz, Hay Ryad, 10100 Rabat, Morocco							TENDRARA PROJECT														
Contractor							Provision of a m-LNG plant for production for ManaEnergy														
ITALFLUID ITALFLUID GEOENERGY Marocco Branch Boulevard de Bourgogne Rue Jaafar Ibn Habib Residence Al Mashrik II 1ere Etage N3 Casablanca																					
Engineering																					
ITALFLUID ITALFLUID GEOENERGY Via Antonio Vivaldi, 4 Montesilvano (PE) Tel:085-4224374 / fax: 085-4220742																					
Facility Name			Location				Scale		Sheet of Sheets												
TENDRARA			MOROCCO				A4		1 of 18												
Document Title					Doc. Code																
P-PK-PK-341-1					ELD01-ING-FR01																
AMMONIA COMPRESSOR PACKAGE 1					Vendor Doc. code																
					IE24090_SE05_P-PK-PK-341-1																
			Project : P-PK-PK-341-1				Customer : ManaEnergy Managem Group				Object :										
General Contractor :		Manufacture : esa esaelettromeccanica.com		AMMMONIA COMPRESSOR PACKAGE 1				Contract : IE24090				Rev. 2		Data : 31/01/2025				Date : 26/03/2024			
								S/N : IE24090_SE05				Rev. 1		Data : 06/12/2024				Sheet : 1		of : 18	

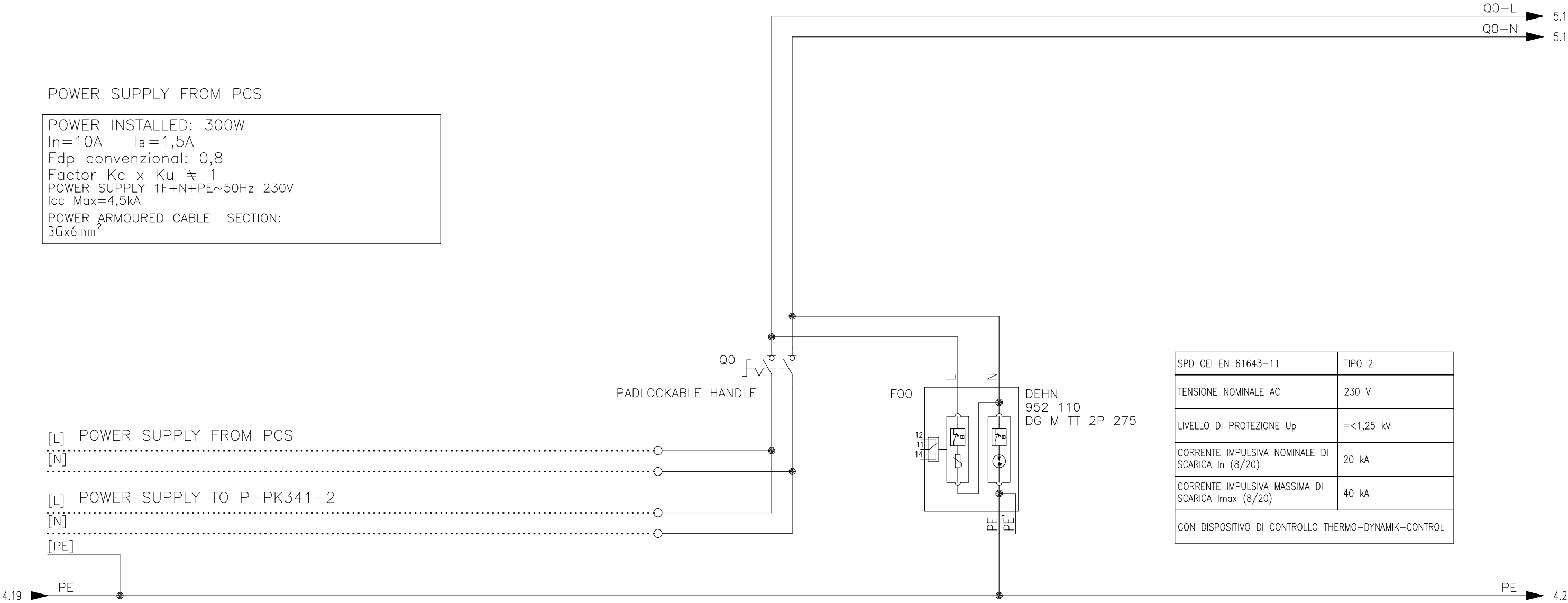
CIRCUIT SYMBOLS ACCORDING TO CEI STANDARD

	ISOLATOR		ELECTRONIC TIMER WITH PICK-UP DELAY		MUSHROOM PUSH-BUTTON PULSE CONTACT WITH STAY-PUT MAKE CONTACT OPERATED BY STRIKING		STEDY OR FLASHING LIGHT
	FUSED ISOLATOR		ELECTRIC PULSE METER		KEY OPERATING DEVICE FOR SELECTORS, PUSH-BUTTON, ETC.		PLC UNIT
	CIRCUIT BREAKER WITH THERMAL AED MAGNETIC OVERCURRENT RELEASES		FUSE CARRIER		PRESSURE SWITCH		CONTROL INSTRUMENT
	CIRCUIT BREAKER WITH MAGNETIC RELEASE		MAKE CONTACT OF CONTACTOR		TEMPERATURE SWITCH		ENERGY METER
	CIRCUIT BREAKER WITH THERMAL, MAGNETIC AND DIFFERENTIAL RELEASES		MAKE CONTACT OF CONTACTOR		FLOW SWITCH		MEASURING INSTRUMENT
	RESIDUAL CURRENT CIRCUIT BREAKER "AC" TYPE FOR ALTERNATING CURRENT "A" TYPE FOR ALTERNATING OR PULSATING CURRENTS		MAKE AND BREAK CONTACT		FLOAT SWITCH		VOLTAGE TRANSFORMER
	THERMAL RELAY (OVERLOAD RELAY)		MAKE AND BREAK CONTACT WITH DROP-OUT DELAY		BUCHHOLZ RELAY		CURRENT TRANSFORMER
	COIL OF CONTACTOR RELAY		MAKE AND BREAK CONTACT WITH PICK-UP DELAY		CAM OPERATED DEVICE		ACOUSTIC SIGNALLING DEVICE
	COIL OF SHUT-OFF ALARM MAKE CONTACT ON ENERGIZATION		MAKE AND BREAK CONTACT WITH DROP-OUT AND PICK-UP DELAY (FLIP-FLOP)		PROXIMITY SWITCH		RESISTANCE
	COIL OF SHUT-OFF ALARM DENERGIZES IN CASE OF UNDERVOLTAGE		MAKE AND BREAK CONTACT OF THERMAL RELAY		TEMPERATURE SENSOR		CAPACITOR
	COIL OF OPERATING DEVICE WITH DROP-OUT DELAY		MAKE AND BREAK CONTACT OF LIMIT SWITCH		PHOTOELECTRIC SENSOR		SINGLE-PHASE ELECTRIC MOTOR
	COIL OF OPERATING DEVICE WITH PICK-UP DELAY		MULTI POSITION SWITCH (GENERAL SYMBOL)		SOLENOID VALVE		THREE-PHASE ELECTRIC MOTOR
	COIL OF OPERATING DEVICE WITH DROP-OUT AND PICK-UP DELAYS		MAKE AND BREAK CONTACT OPERATED BY TURNING WITH SPRING RETURN		PHOTOCELL		
	COIL OF LATCHING RELAY (STEP-BY-STEP FUNCTION)		MAKE AND BREAK CONTACT OPERATED BY PUSHING WITH SPRING RETURN				
	ELECTRONIC OR ELECTRO-MECHANICAL TIME SWITCH		MAKE AND BREAK CONTACT OPERATED BY PULLING WITH OR WITHUOT SPRING RETURN				

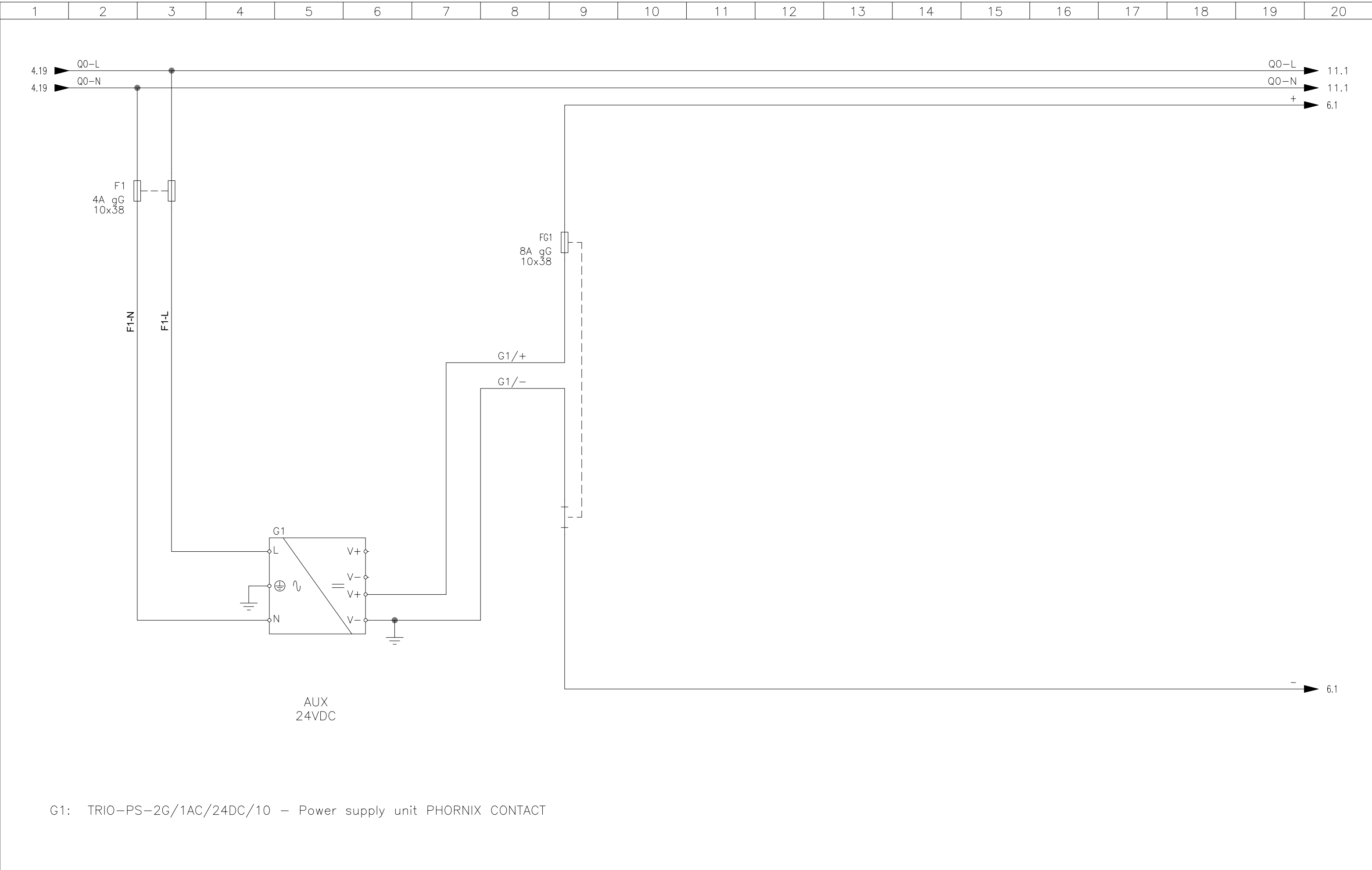
CONDUCTOR COLOR CODING INSIDE CONTROL PANEL	
AC AND DC POWER CIRCUIT	BLACK
NEUTRAL CONDUCTOR OF POWER CIRCUIT	LIGHT-BLUE
PROTECTIVE CONDUCTOR (EARTH CONDUCTOR)	YELLOW-GREEN
110/220V AC AUXILIARY CIRCUIT (*)	RED
24V AC AUXILIARY CIRCUIT	WHITE
24V AC AND DC PLC INPUT	GRAY
24V AC PLC OUTPUT	WHITE
24V AC AND DC INTERLOCKING CIRCUIT	ORANGE
24V DC AUXILIARY CIRCUIT AND PLC OUTPUT	DARK-BLUE
INTRINSIC SAFE SIGNAL	BLUE
(*) : RED FOR 110V AND BROWN FOR 220V, IN CASE BOTH VOLTAGE ARE USED	

CODE COLOUR	
COLOUR	CODE LETTERS
BLACK	BK
BROWN	BN
RED	RD
ORANGE	OG
YELLOW	YE
GREEN	GN
BLUE	BU
GREY	GY
WHITE	WH
SILVER	SR
CLEAR	CL
GREEN AND YELLOW	GNYE

	POWER MAIN DISCONNECT SWITCH																FEEDER ID		
	1F+N+PE~50Hz 230V																RATED POWER Pn~RATED VOLTAGE Vn		
																	RATED CURRENT		

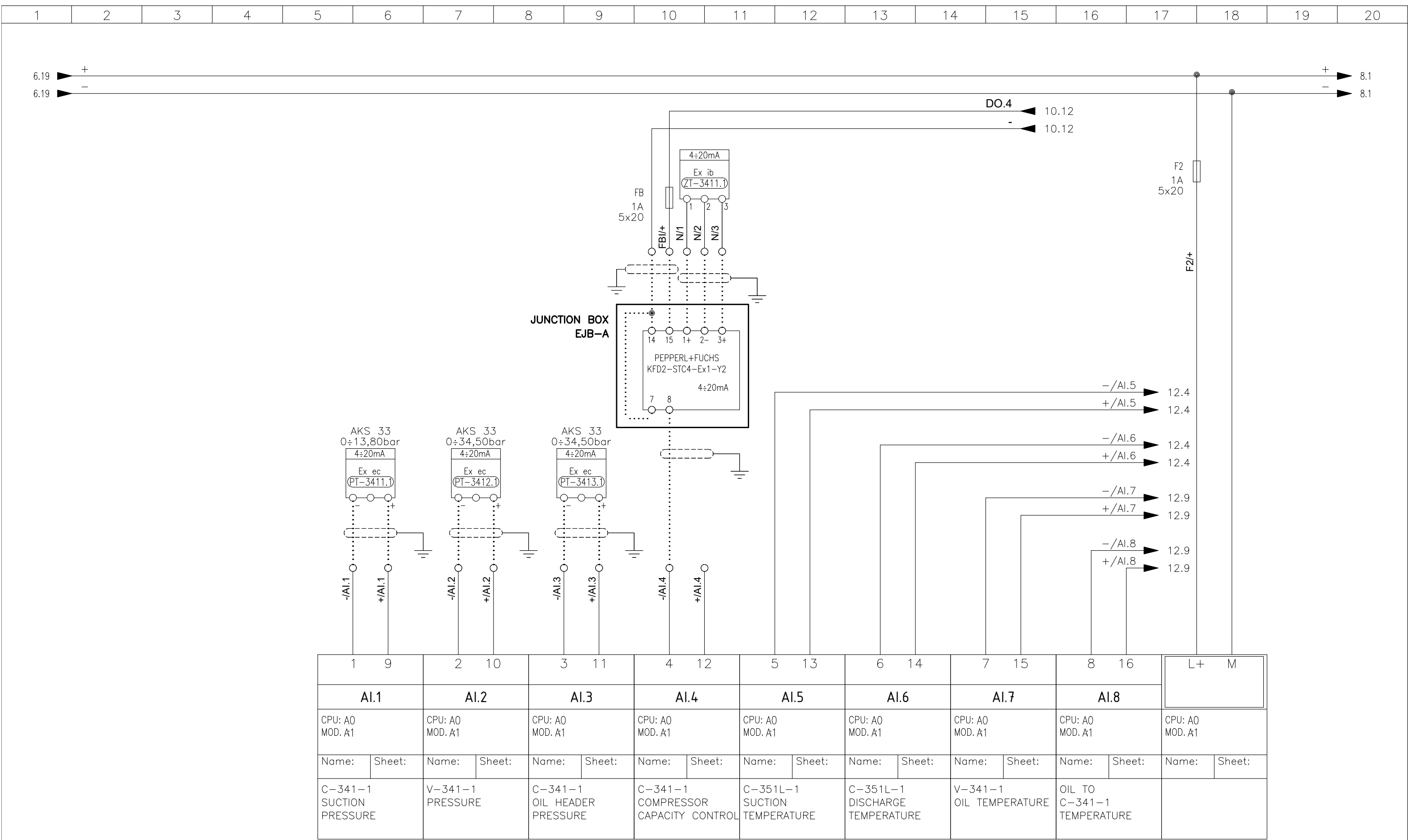


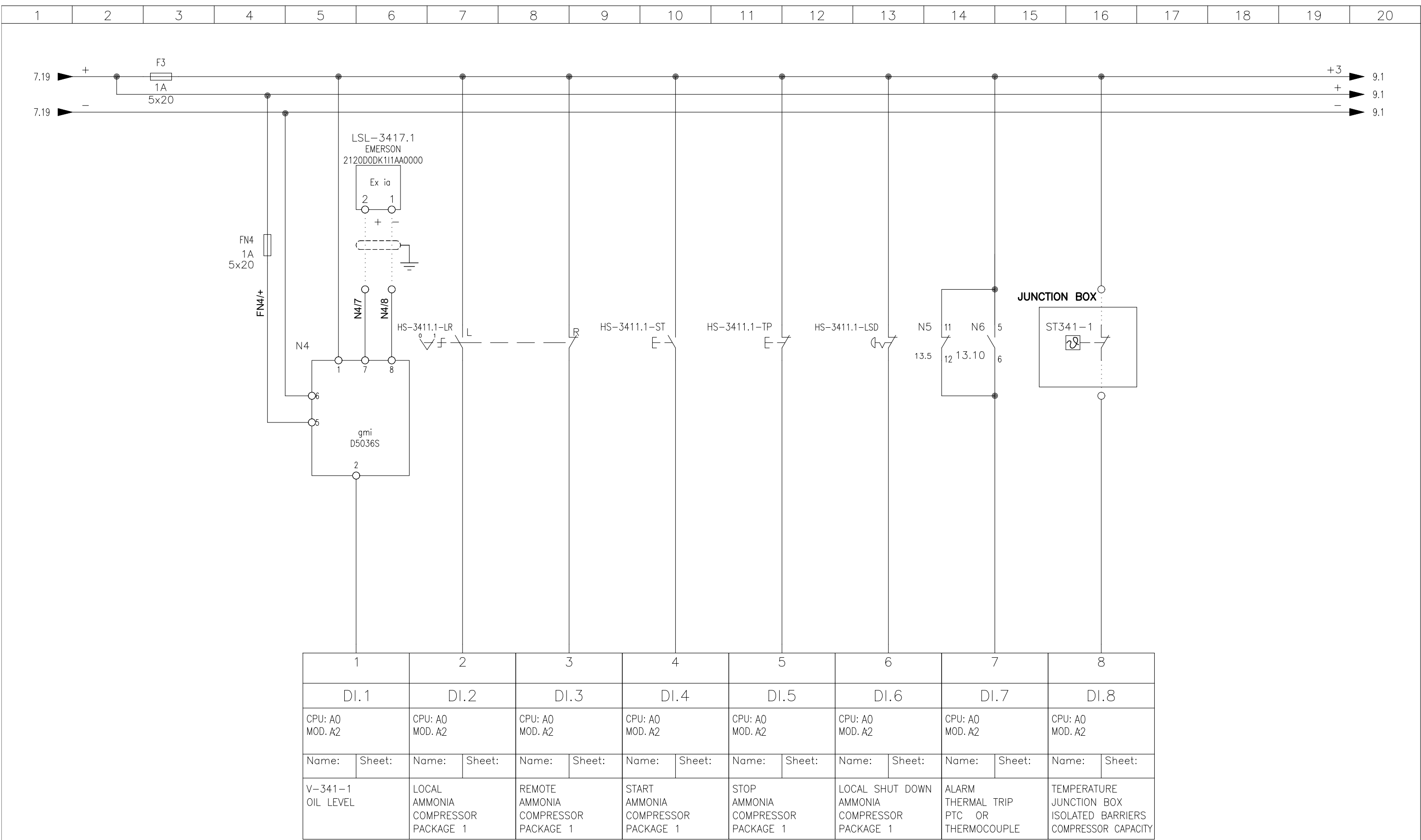
CONDUCTOR PROTECTION	TECHNOR PSRC-220																TYPE		CONDUCTOR PROTECTION
	20A																RATED BREAKING CAPACITY Icu		
																	RATED or THERMAL CURRENT In or Ith		
																	MAGNETIC RELEASE CHARACTERISTICS Im		
																	RATED R.C. SENSITIVITY Idn		
																	WIRE SECTION		
																	CABLE TYPE		
																	CABLE CARRYING LOAD Iz		
																	CABLE LENGHT		
																	LINE ID		
WIRE ID ON TERMINAL																			

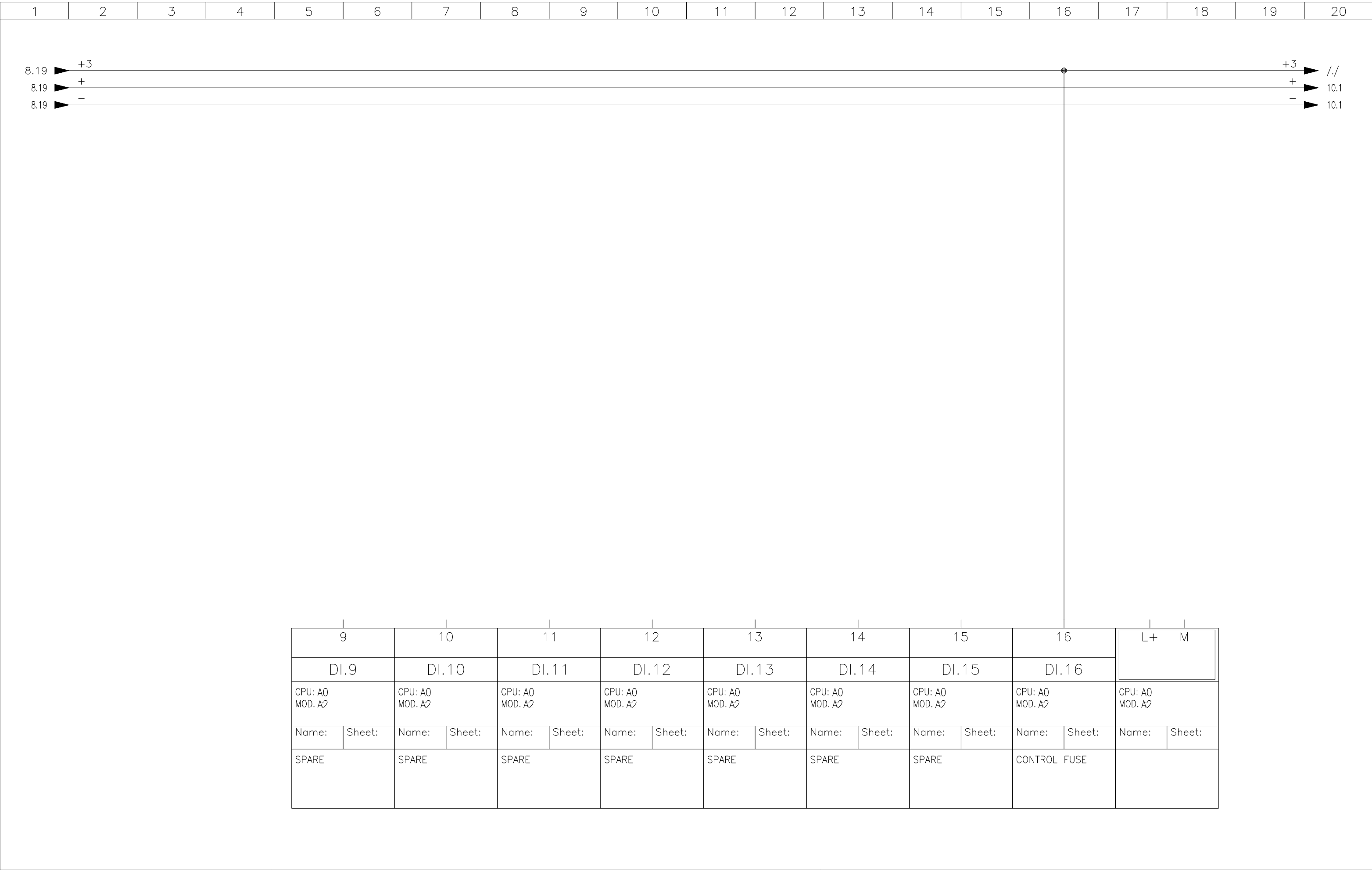


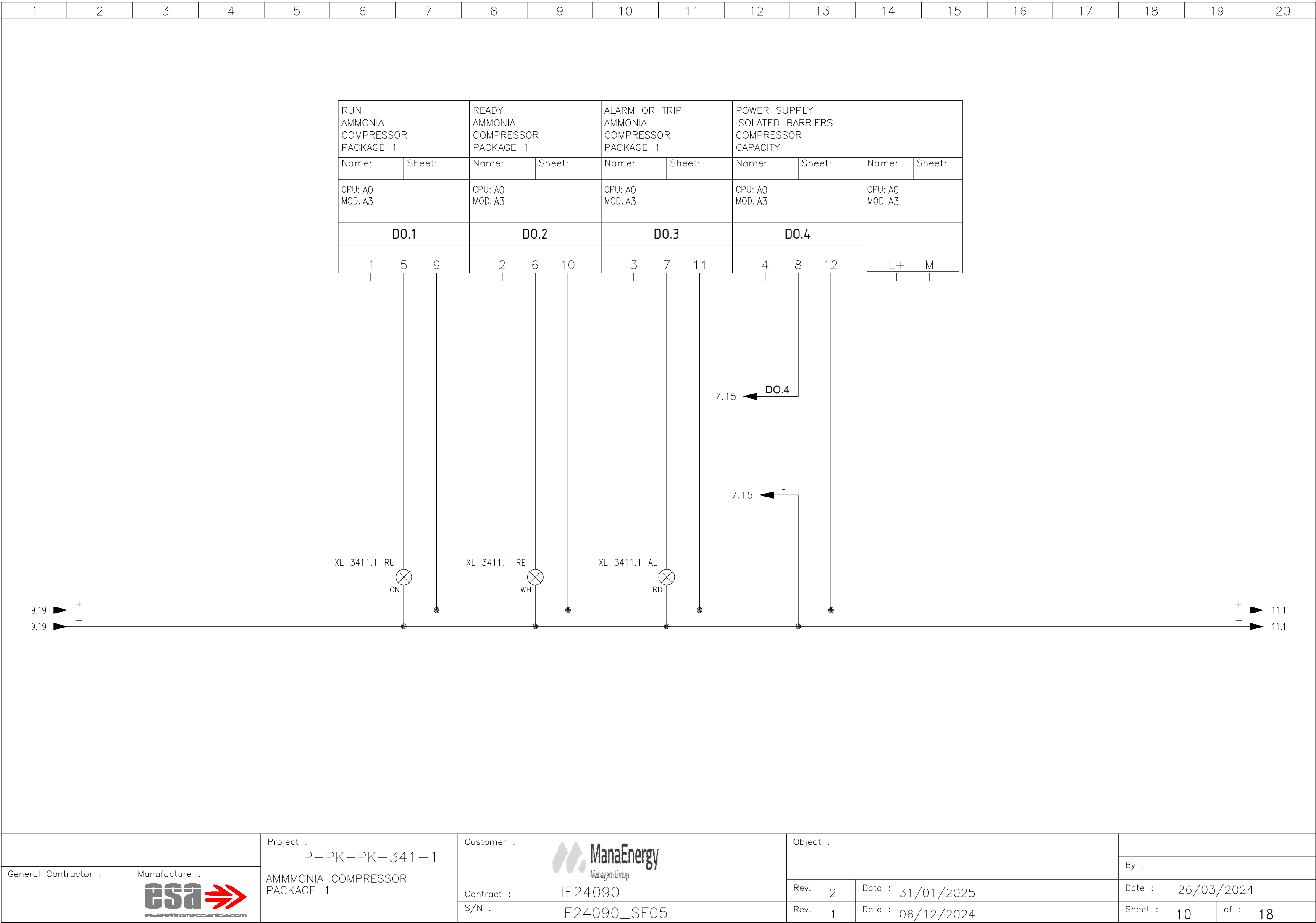
G1: TRIO-PS-2G/1AC/24DC/10 - Power supply unit PHORNIX CONTACT

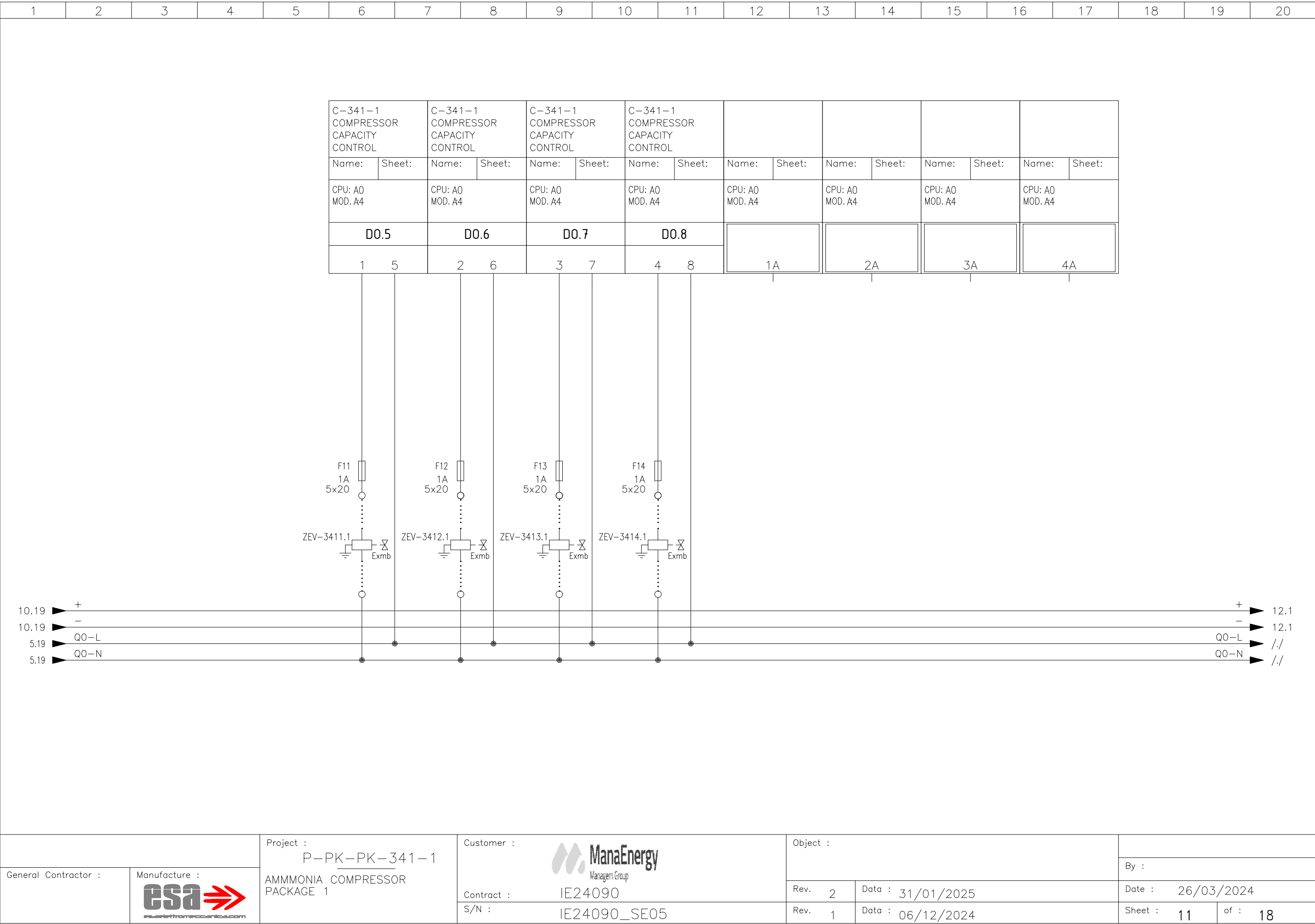
General Contractor : Manufacture : 		Project : P-PK-PK-341-1 AMMMONIA COMPRESSOR PACKAGE 1	Customer : Contract : IE24090 S/N : IE24090_SE05		Object : Rev. 2 Data : 31/01/2025 Rev. 1 Data : 06/12/2024		By :	
							Date : 26/03/2024	Sheet : 5 of : 18

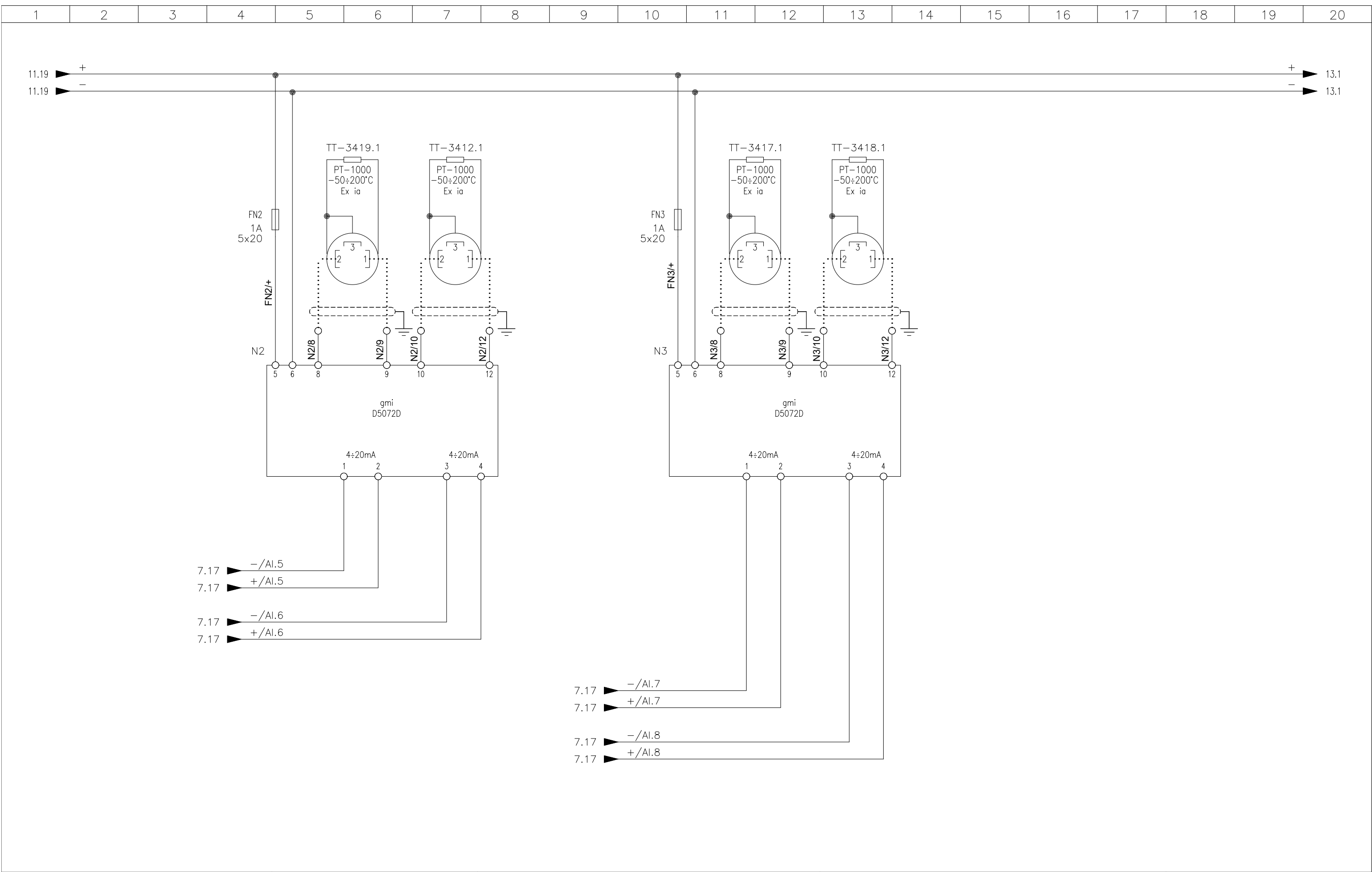












TT-3417.1

PT-1000

-50÷200°C

Ex ia

TT-3418.1

PT-1000

-50÷200°C

Ex ia

FN3

1A

5x20

FN3/+

N3

5

6

8

9

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12

gmi

D5072D

4÷20mA

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2

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4

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7.17

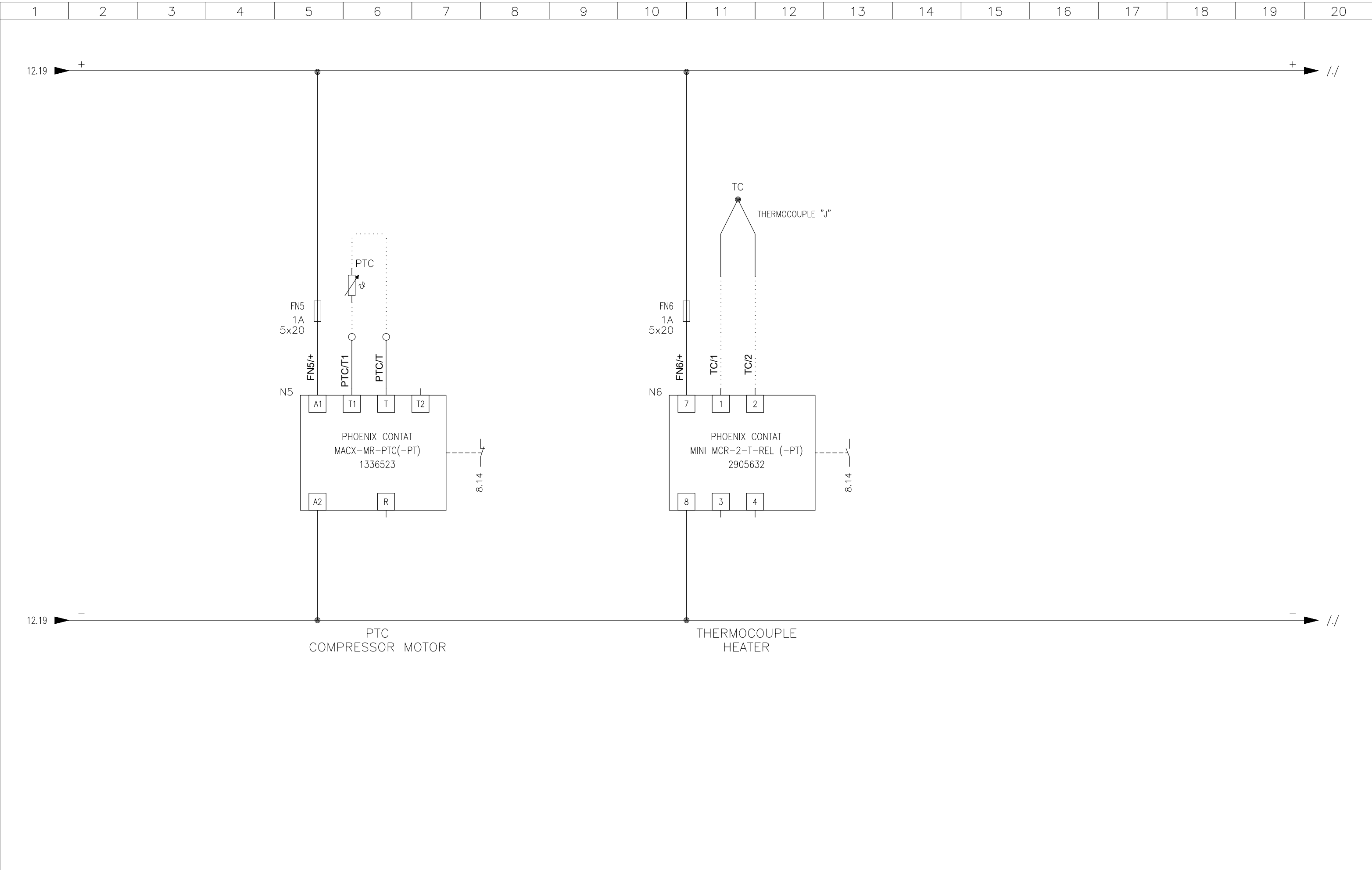
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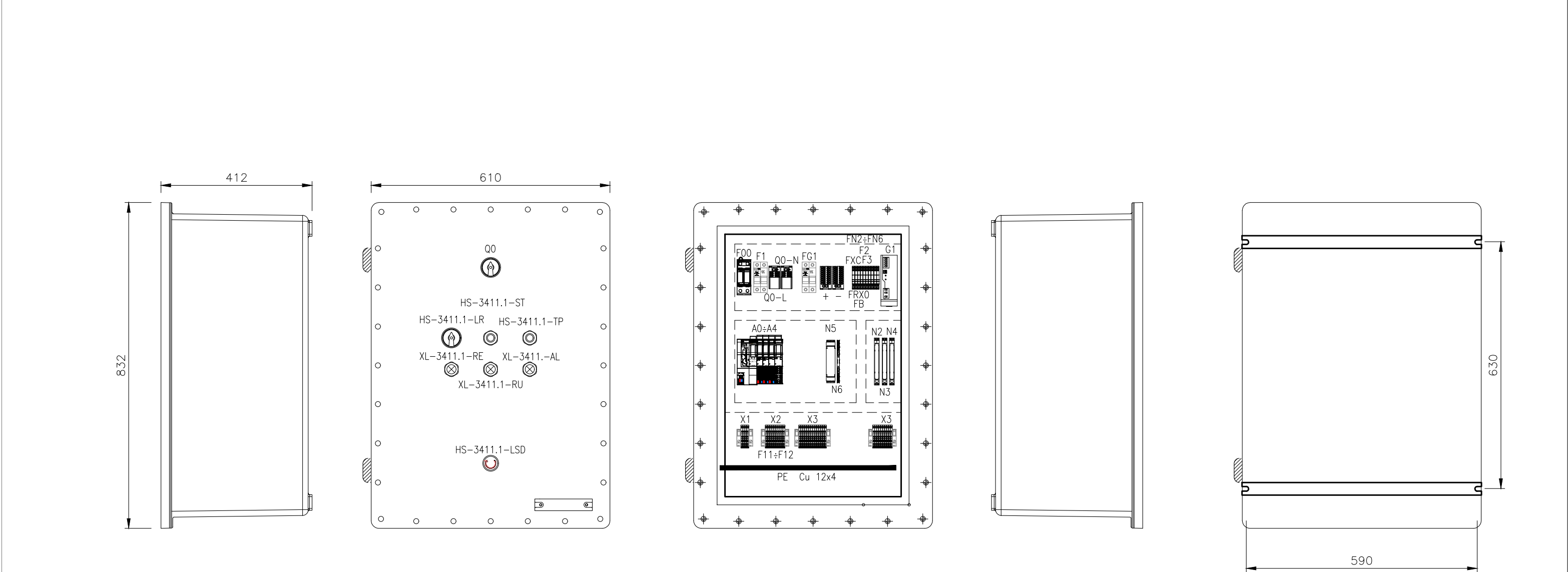


PTC

COMPRESSOR MOTOR

THERMOCOUPLE

HEATER



MILANO-ITALY Tel 024455358 Fax 0248401741

Type: EJB-H
230Vac/50Hz 24Vdc 10A
Max Dis. P: 81 W
Year s.n 25/CMC/0005

CE 0722 **Ex II 2GD**
INERIS18ATEX0026X/00
Ex db [ia Ga] IIB T6
Ex tb [ia Da] IIC T85 °C IP66
-20°C < Tamb < +50°C

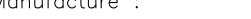
Cable entries: see instructions

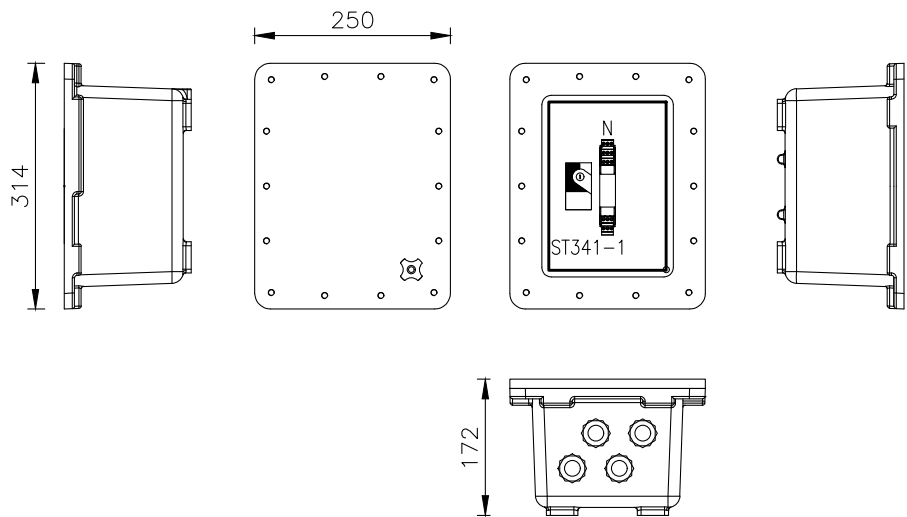
WARNINGS: DO NOT OPEN WHILE ENERGIZED
DO NOT OPEN IF AN EXPLOSIVE ATMOSPHERE MAY BE PRESENT
READ INSTRUCTIONS NOTE BEFORE INSTALLATION AND USE

General Contractor :		Manufacture : 	Project : P-PK-PK-341-1 AMMMONIA COMPRESSOR PACKAGE 1	Customer :  Contract : IE24090 S/N : IE24090_SE05	Object :		By :	
					Rev. 2	Data : 31/01/2025	Date : 26/03/2024	
					Rev. 1	Data : 06/12/2024	Sheet : 14	of : 18

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
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INITIALS	SUPPLIER PN	DESCRIPTION	SUPPLIER
–	3273110	PTFIX 6/18X2,5–NS35 GY – Distribution block	PHOENIX CONTACT
+	3273110	PTFIX 6/18X2,5–NS35 GY – Distribution block	PHOENIX CONTACT
A0	6AG1155–6AU01–7CN0	SIPLUS ET 200SP IM155–6PN HF interface module	SIEMENS
A0	6AG1193–6AG20–2AA0	ET 200SP BA LC/RJ45	SIEMENS
A1	6AG1134–6GF00–7AA1	AI 8xI 2–/4–wire BA	SIEMENS
A1	6AG1193–6BP00–7DA0	ET 200SP BU15–P16+A0+2D push–in terminals	SIEMENS
A2	6AG1131–6BH01–7BA0	ET 200SP DI 16x24 V DC	SIEMENS
A2	6AG1193–6BP00–7BA0	ET 200SP BU15–P16+A0+2B push–in terminals	SIEMENS
A3	6AG1132–6GD51–7BA0	ET 200SP RQ 4x24VDC/2A CO ST	SIEMENS
A3	6AG1193–6BP00–7BA0	ET 200SP BU15–P16+A0+2B push–in terminals	SIEMENS
A4	6AG1132–6HD01–7BB1	ET 200SP RQ 4x 120V DC..230VAC/5A ST	SIEMENS
A4	6AG1193–6BP20–7BB0	ET 200SP BU20–P12+A4+0B push–in terminals	SIEMENS
BOX O.F.	LOGILINK FB5000	Fibre optic DIN rail splice box for LC or SC couplers	REICHELT
F00	952110	DG M TT 2P 275 – SPD	DEHN
F1	2301138	Fuseholder for cylindrical fuses – 1P+N for fuses 10,3x38mm – BCH 1x38N 32A 690V	ITALWEBER
F11	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
F12	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
F13	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
F14	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
F2	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
F3	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
FB	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
FG1	2302038	Fuseholder for cylindrical fuses – 2P for fuses 10,3x38mm – BCH 2x38 32A 690V	ITALWEBER
FN2	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
FN3	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
FN4	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
FN5	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
FN6	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
FRK0	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
FXC	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
G1	2903149	TRIO–PS–2G/1AC/24DC/10 – Power supply unit	PHOENIX CONTACT
HS–3411.1–LR	PSRC–D220	CHANGE OVER SWITCH	ITALWEBER
HS–3411.1–LSD	PLC–S	MUSHROOM HEAD MAINTAINED (TWIST TO RELEASE)	CMC
HS–3411.1–LSD	E20GE11	CONTACT BLOCK N.O. +N.C. WITH FLANGE	ELFI
HS–3411.1ST	PLA30	PUSH BUTTON GREEN	CMC
HS–3411.1ST	E20GE11	CONTACT BLOCK N.O. +N.C. WITH FLANGE	ELFI
HS–3411.1TP	PLA20	PUSH BUTTON RED	CMC
HS–3411.1TP	E20GE11	CONTACT BLOCK N.O. +N.C. WITH FLANGE	ELFI
N2	D5072D	I.S. SIL2 Multifunction Temperature Converter	GMI
N3	D5072D	I.S. SIL2 Multifunction Temperature Converter	GMI
N4	D5036S	I.S. SIL2 Switch/Proximity Relay–Out Repeater	GMI
N5	1336523	MACX–MR–PTC – Monitoring relay	PHOENIX CONTACT
N6	2905632	MINI MCR–2–T–REL – Limit value switch	PHOENIX CONTACT
P–PK–341–1	EJB–H	ENCLOSURE 832x612x412 mm L x H x D	CMC
Q0	PSRC–320	SELECTOR SWITCH	CMC
Q0–L	3273066	PTFIX 6/6X2,5–NS35 GY – Distribution block	PHOENIX CONTACT
Q0–N	3273066	PTFIX 6/6X2,5–NS35 GY – Distribution block	PHOENIX CONTACT
XL–3411.1	PLD30	PILOT LAMP GREEN	CMC
XL–3411.1	020PD9	LAMPHOLDER BA9S, DIRECT SUPPLY	ELFI
XL–3411.1–AL	PLD20	PILOT LAMP RED	CMC
XL–3411.1–AL	020PD9	LAMPHOLDER BA9S, DIRECT SUPPLY	ELFI
XL–3411.1–RE	PLD10	PILOT LAMP WHITE	CMC
XL–3411.1–RE	020PD9	LAMPHOLDER BA9S, DIRECT SUPPLY	ELFI

General Contractor :		Manufacture : 	Project : P-PK-PK-341-1 AMMMONIA COMPRESSOR PACKAGE 1	Customer :  Management Group	Object :		By :	
					Rev. 2	Data : 31/01/2025	Date : 26/03/2024	
				Contract : IE24090	Rev. 1	Data : 06/12/2024	Sheet : 15	of : 18
				S/N : IE24090_SE05				



INITIALS	SUPPLIER PN	DESCRIPTION	SUPPLIER
EJB-A	EJB-A	ENCLOSURE 250x314x172 mm L x H x D	CMC
N	KFD2-STC4-Ex1-Y2	SMART Transmitter Power Supply	PEPPERL+FUCHS
ST341-1	TMS NC	THERMOSTAT	O.ERR

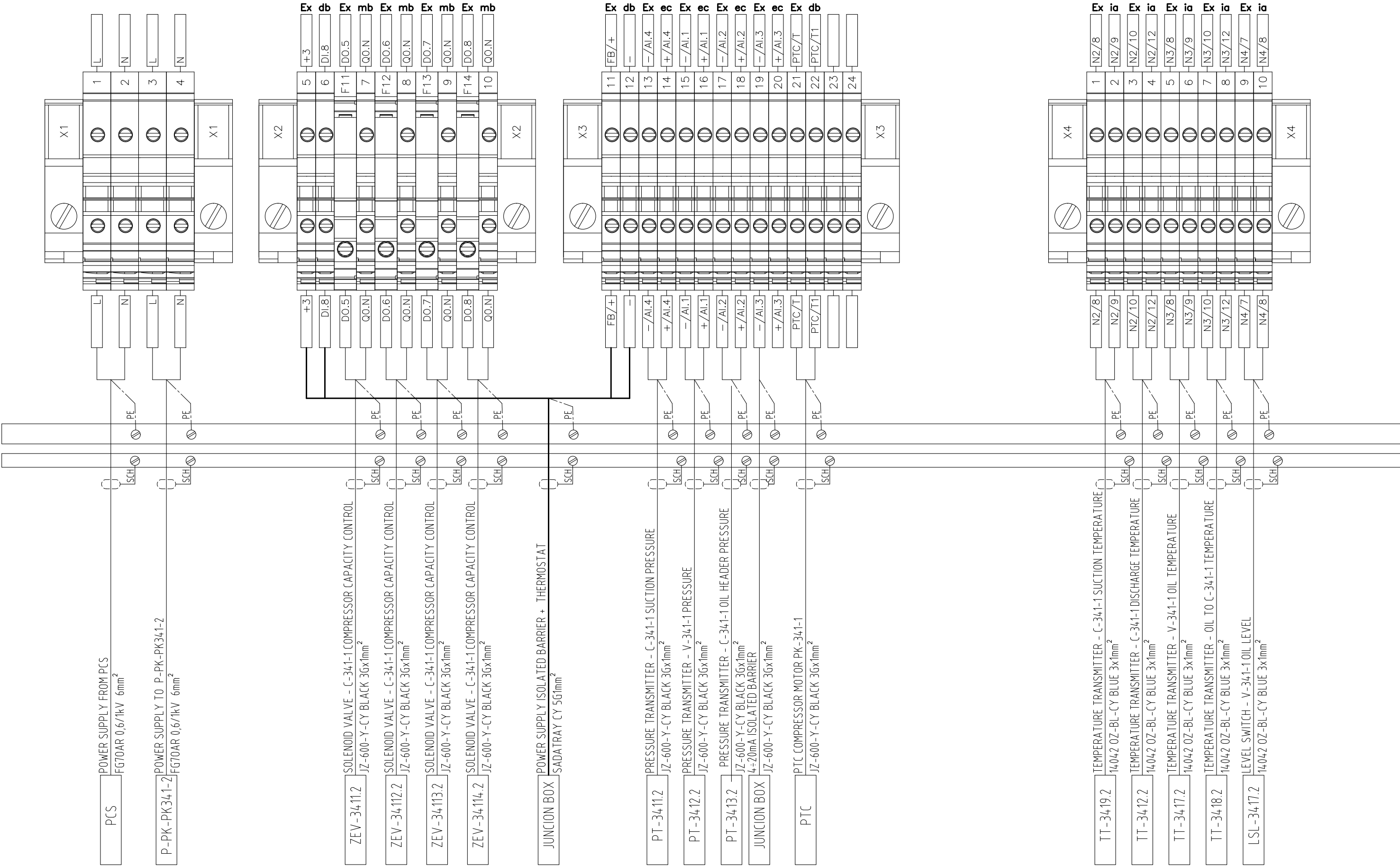


MILANO-ITALY Tel 024455358 Fax 0248401741

Type: EJB-A
24Vdc 10A
Max Dis. P: 6 W
Thermal probe: 55±5 °C
Year s.n 25/CMC/0010

CE 0722 Ex II 2GD
INERIS18ATEX0026X/00
Ex db [ia Ga] IIB T6
Ex tb [ia Da] IIC T85 °C IP66
-20°C < Tamb < +50°C

Cable entries: see instructions
WARNINGS: DO NOT OPEN WHILE ENERGIZED
DO NOT OPEN IF AN EXPLOSIVE ATMOSPHERE MAY BE PRESENT
READ INSTRUCTIONS NOTE BEFORE INSTALLATION AND USE



INTRINSIC CIRCUIT INSIDE

TYPE OF PROTECTION	
FLAMEPROF ENCLOSURE	EX db
INCREASED SAFETY	EX ec
INTRINSIC SAFETY	EX ia
ENCAPSULATRION	EX mb

		Project : P-PK-PK-341-1	Customer : 		Object :			
General Contractor :	Manufacture : 	AMMMONIA COMPRESSOR PACKAGE 1	Contract : IE24090		Rev. 2	Data : 31/01/2025	By : Date : 26/03/2024	
			S/N : IE24090_SE05		Rev. 1	Data : 06/12/2024	Sheet : 18	of : 18